# guardrails.py - Hard Safety Rules



## What this file does (big picture)

guardrails.py is the agent's non-negotiable policy layer.

Business rules encoded here:

1) Do not diagnose medical conditions
2) Do not recommend changing prescriptions
3) Escalate emergencies
4) Do not promise refunds
5) Do not expose another customer's order information

Two checkpoints: check_input (before we answer) and check_output (after a draft exists - including future AI drafts).

Professional habit: put safety in its own module so every channel (CLI today, Twilio later) can call the same rules.

## New words you will see in this file

| | |
|---|---|
| **guardrail** | A forced safety check that overrides clever or unsafe replies. |
| **allow / block / escalate** | The three actions: proceed, refuse safely, or hand to a human. |
| **input check** | Inspect the customer's message before classification. |
| **output check** | Inspect our draft reply before the customer sees it. |
| **regex** | A text pattern (re.search) used to catch refund-promise wording. |
| **@property** | A method used like a field - result.allowed reads as a True/False flag. |

### Flow while the program runs

check_input(question) -> allow, block, or escalate.

If allow: build draft elsewhere -> check_output(draft) / apply_output_guardrails.

Blocked drafts are replaced with a safe message; emergencies escalate first.

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**1-17**

```
"""Guardrails for the sleep-company AI agent..."""
```

**Plain English:**
Module docstring lists the five business rules and explains input vs output checks.

**19-23**

```
imports: re, dataclass, Optional
```

**Plain English:**
Regex for pattern matching, dataclass for the result object, Optional for missing reasons.

**25-28**

```
ALLOW / BLOCK / ESCALATE = "allow" / "block" / "escalate"
```

**Plain English:**
Action constants so main.py compares names, not magic strings scattered everywhere.

**30-36**

```
REASON_* constants
```

**Plain English:**
Machine-readable why-codes for logs and future EscalationPacket notes.

**39-56**

```
@dataclass class GuardrailResult
```

**Plain English:**
Traffic-light result: action, optional reason, customer-facing message. .allowed is True only for allow.

**59-61**

```
def _contains_any(...)
```

**Plain English:**
Private helper: does the text contain any clue phrase?

**66-77**

```
_EMERGENCY_CLUES list
```

**Plain English:**
Phrases that mean stop and escalate (chest pain, suicidal language, call 911, etc.).

**79-102**

```
_DIAGNOSIS_CLUES and _PRESCRIPTION_CLUES
```

**Plain English:**
Phrases that mean refuse: diagnosing conditions or advising medication changes.

**104-121**

```
_OTHER_CUSTOMER_CLUES
```

**Plain English:**
Privacy: looking up someone else's order (husband/wife/friend) is blocked.

**124-180**

```
def check_input(question)
```

**Plain English:**
Order matters: emergency first, then prescription, diagnosis, other-customer privacy. Otherwise allow.

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**185-208**

```
refund / diagnosis / prescription output patterns
```

**Plain English:**
Patterns and clues that mean a *draft reply* broke policy (especially refund promises).

**211-273**

```
def check_output(draft, question)
```

**Plain English:**
Scan the draft for refund promises, diagnosis/prescription wording, and unexpected emails not present in the question.

**276-281**

```
def apply_output_guardrails(...)
```

**Plain English:**
Convenience wrapper: return the draft if safe, else the safe refusal message.

**284-293**

```
def policy_summary()
```

**Plain English:**
Printable list of active rules for operators and documentation.